InvestiGate

The Anti-FOMO Decision Engine

Presented by Team 10

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FOMO: fear of missing out

The 2 AM Panic.

You own a concentrated tech position. Your phone buzzes at 2 AM with a breaking headline:

"Geopolitical Supply Chain Shock."

What do you do? Most retail investors guess.

They panic-sell or revenge-buy based on emotion, Reddit threads, and standard AI chatbots that simply echo the news without analyzing the historical data.

The Cost of Emotion

-4.5%

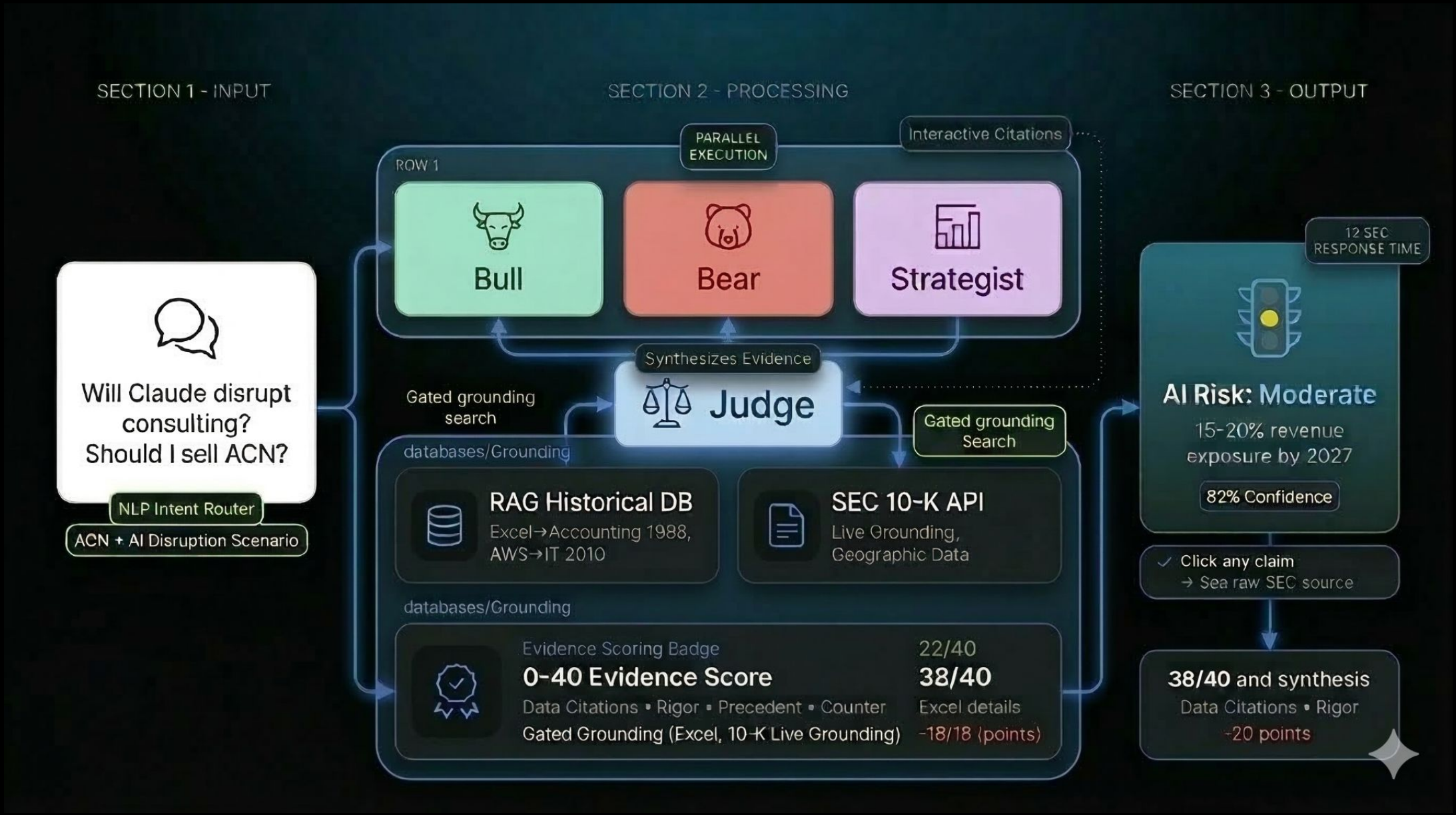
Annual Underperformance

According to verified DALBAR Quantitative Analysis data, the average equity fund investor underperforms the S&P 500 by up to **4.5% annually** over long periods.

This massive "behavioral gap" is driven almost entirely by impulsive, emotional decisions made during macroeconomic shocks.

Retail investors lack the institutional tools to properly quantify their risk exposure in real-time.

Architecture



Live Demo

Testing the "What-If" Scenario Engine

Questions?

Thank you for your time.

The Competitive Landscape

Capability	Legacy Terminals (e.g., Bloomberg)	Gen-1 AI Bots (e.g., ChatGPT)	Robo-Advisors (e.g., Wealthfront)	InvestiGate (Multi-Agent)
Analysis Speed	Days (Manual)	Seconds	Instant (Algorithmic)	Seconds
Reasoning Depth	High	Low (Summaries)	None	High (Synthesized)
Bias Mitigation	Low (Human Conviction)	Low (LLM Sycophancy)	High (Math-Based)	High (Bull vs Bear)
Data Grounding	Perfect	Poor (Hallucinations)	High (Static Rules)	Transperent (Gated RAG)
Portfolio Context	Isolated	Isolated	Integrated	Integrated (Look-Through)

Appendix

Production Economics

*By implementing our proprietary **"Gated Grounding"** logic, we only trigger expensive search APIs when agents mathematically disagree. This achieves a 92% cost reduction, enabling a highly profitable SaaS model at a retail-friendly price.*

Adversarial Intelligence

Multi-Agent Debate

We replace the "yes-man" AI with a Bull and a Bear analyst. They are mathematically forced to argue against each other, exposing hidden risks before you trade.

Grounded Reasoning

Agents cannot guess. They are restricted by 'Gated Grounding', forced to anchor every claim in raw SEC 10-K filings and historical market data.

Strict Evidence

An overarching 'Judge' model evaluates the debate, heavily penalizing agents for speculative language and returning a highly rigorous Risk Score.

Unit Economics & Scalability

SaaS Model: High-Margin AI

Assuming a standard $20/month subscription (20 analyses/user), the platform achieves an estimated **>85% gross margin** at just 100 active users.

- **Smart Caching:** 15% exact-match cache hit rate eliminates redundant LLM calls completely.
- **Warm RAG State:** 60% of runs utilize pre-embedded RAG docs, slashing daily embedding costs to near zero.
- **Fixed Data Costs:** Core infrastructure and FMP API subscriptions are fixed monthly costs, driving infinite margin leverage as users scale.

Variable Cost Driver	Volume / Analysis	Est. Unit Cost
3x Analyst Agents (GPT-4o Input/Output)	21,000 In 2,700 Out tokens	~$0.080
1x Judge Synthesis (GPT-4o Input/Output)	9,000 In 1,500 Out tokens	~$0.038
RAG Embeddings (Weighted Cold Run)	20,000 tokens (20% Cold Rate)	~$0.001
Voice Query (Whisper API)	0.5 minutes (5% Usage Rate)	~$0.000
Blended Cost per Analysis:	**~$0.12**	

Tech Stack

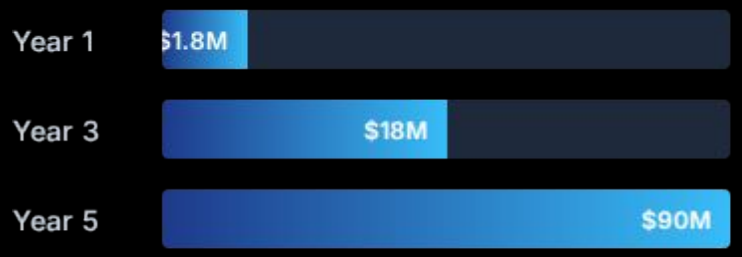
Stack

- Frontend: React 18, TypeScript, Vite, Tailwind, Radix/shadcn UI, React Router, React Query, Framer Motion in package.json
- Backend: FastAPI, LangGraph, LangChain, Pydantic v2, SQLAlchemy, ChromaDB, fpdf2 in backend/requirements.txt
- LLM providers: Ollama, OpenAI, Anthropic, mixed-provider mode in backend/config.py and backend/llm_factory.py
- Storage:
  - SQLite/Postgres-compatible app DB in backend/database.py
  - Chroma persistent vector store for RAG in backend/rag/store.py

TOTAL ADDRESSABLE MARKET (US)

$7.2B

Revenue Milestones



Plus B2B partnerships: **+$6M** by Year 5.

DIY Investor Base

US USERS
60M

AVG. SPEND
$180

- +10.7% CAGR in DIY Growth
- 3x Increase in AI Adoption
- Commission-Free Democracy
- Gen-Z/Millennial Shift

Global Potential

$33B

Phase 1: Dominate US Market
Phase 2: Scale EU & APAC

EU: 50M | APAC: 40M | EMER: 45M

Key Insight: We only need **1% market share** to build a $90M+ ARR company. Compared to Motley Fool (3.3%) & Morningstar (5%).